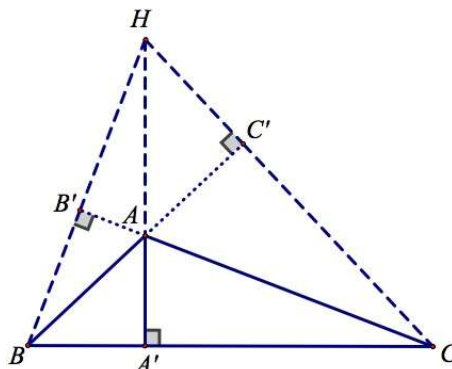
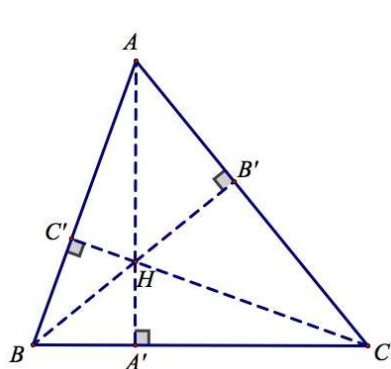


Olympiad Geometry – V

3. Exploring the Triangle

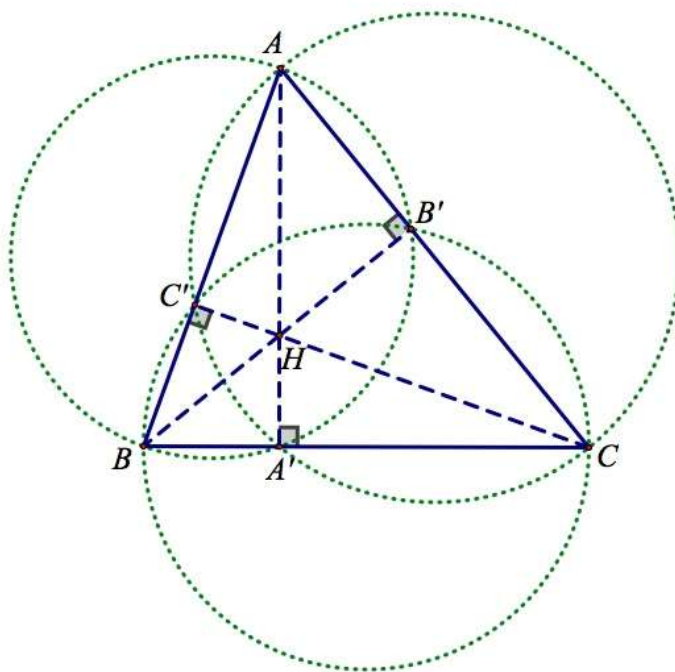
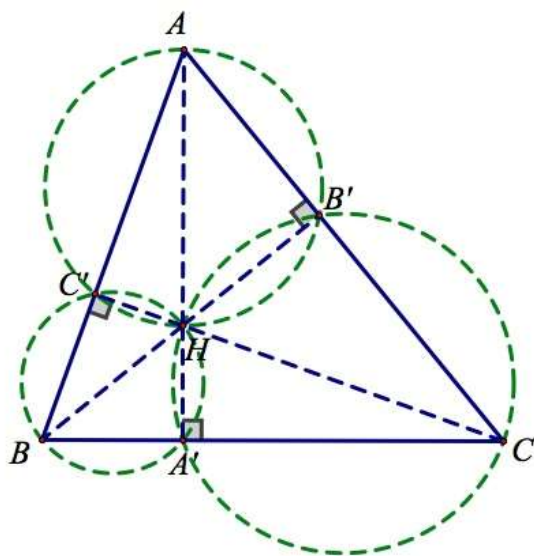
Basic Properties of the Orthocentre

Let AA' , BB' and CC' be the altitudes in triangle ABC with orthocentre H and circumradius R .



Then:

- Quadrilaterals (possibly in different order of vertices) $BCB'C'$, $CAC'A'$, $ABA'B'$ are cyclic with sides BC , CA , AB respectively, as their diameters.
- Quadrilaterals (possibly in different order of vertices) $AC'HB'$, $BA'HC'$, $CB'HA'$ are cyclic with segments AH , BH , CH respectively, as their diameters.
- If angles B and C are acute, then $\angle BHC = 180^\circ - A$ and otherwise $\angle BHC = A$.
- The circumradii of triangles BHC , CHA , AHB are all equal to R .
- Triangles $AB'C'$, $A'BC'$, $A'B'C$ are all similar to triangle ABC with ratios of similitude equal to $|\cos A|$, $|\cos B|$, $|\cos C|$ respectively.
- $AH = 2R|\cos A|$, $BH = 2R|\cos B|$, $CH = 2R|\cos C|$.



Reflection of the Orthocentre

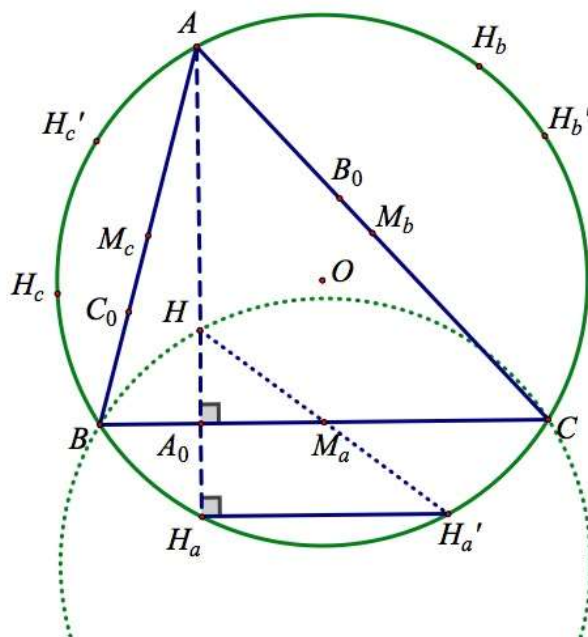
Let ABC be a triangle with orthocentre H . Denote by H_a the reflection of H over the side BC and denote by H_a' the image of H under reflection about the midpoint of BC .

Define points H_b, H_b', H_c, H_c' , analogously. Then points $H_a, H_a', H_b, H_b', H_c, H_c'$ lie on the circumcircle S of triangle ABC and AH_a', BH_b', CH_c' are its diameters.

In other words, the circumcircle of $\triangle BHC$ is the reflection of the circumcircle of $\triangle ABC$ about BC .

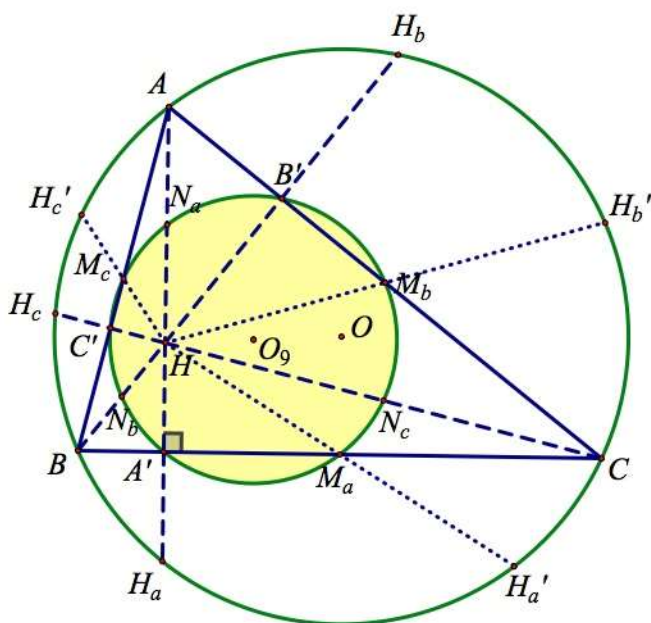
Moreover, we have

$$AH \times HA_0 = BH \times HB_0 = CH \times HC_0$$



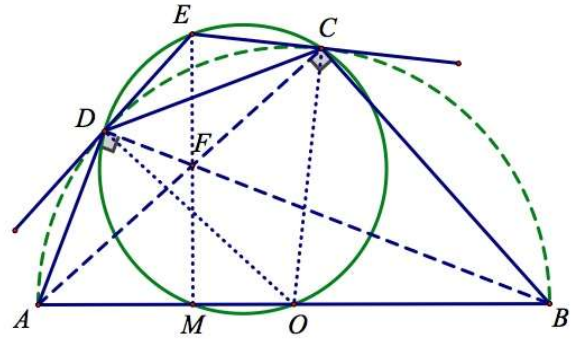
Theorem (Nine-Point Circle)

Let AA', BB', CC' be the altitudes in triangle ABC with orthocentre H , circumcentre O and circumradius R . Denote by M_a, M_b, M_c the midpoints of the sides BC, CA, AB respectively, and let N_a, N_b, N_c be the midpoints of the segments AH, BH, CH respectively. Then points $M_a, M_b, M_c, A', B', C', N_a, N_b, N_c$ lie on a circle with radius $\frac{R}{2}$. The centre O_9 of this circle bisects the segment OH . Segments N_aM_a, N_bM_b, N_cM_c are diameters of the circle.



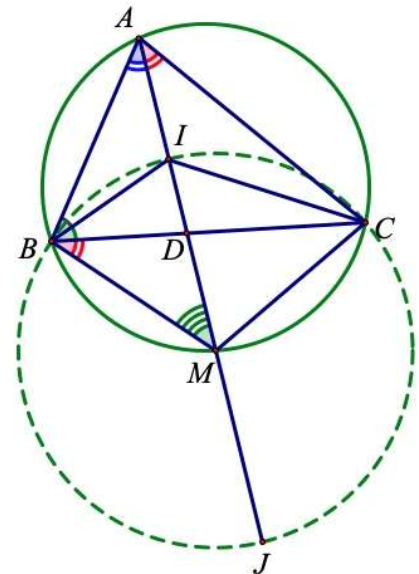
Example 3.1 (China Western MO 2010)

Quadrilateral $ABCD$ is inscribed in a semicircle with diameter AB and centre O . Lines tangent to the semicircle at points C and D meet at E and the segments AC and BD meet at F . Denote by M the intersection of EF and AB . Prove that E , C , M and D are concyclic.

**Basic Properties of the Incentre**

In triangle ABC inscribed in a circle S , let I be the incentre, J the A -excentre, M the midpoint of arc BC of S that does not contain A , and D the foot of the A -angle bisector. Then:

- (a) $\angle BIC = 90^\circ + \frac{1}{2}A$
- (b) M lies on the angle bisector of $\angle A$ and $MB = MC = MI$.
- (c) $\frac{AI}{ID} = \frac{b+c}{a}$
- (d) A, I, M, J are collinear with M being the midpoint of IJ .



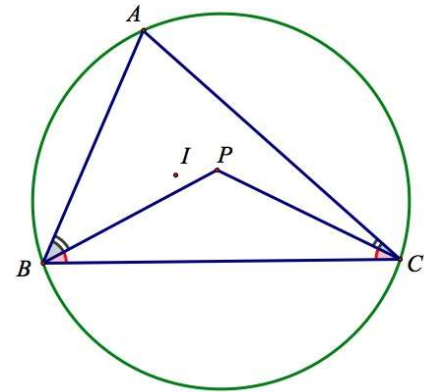
Remark: In our culture, (b) and (d) together called "雞爪定理".

Example 3.2 (IMO 2006 #1)

Let ABC be a triangle with incentre I . A point P in the interior of the triangle satisfies

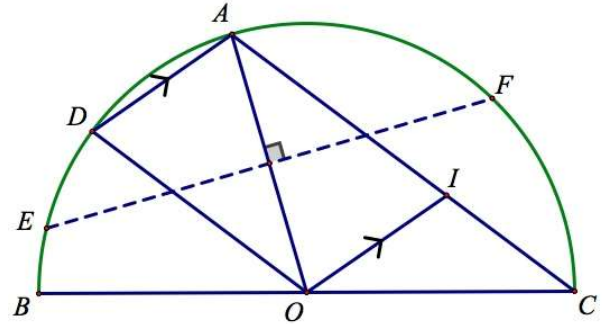
$$\angle PBA + \angle PCA = \angle PBC + \angle PCB.$$

Show that $AP \geq AI$, and that equality holds if and only if $P = I$.



Example 3.3 (IMO 2002 #2)

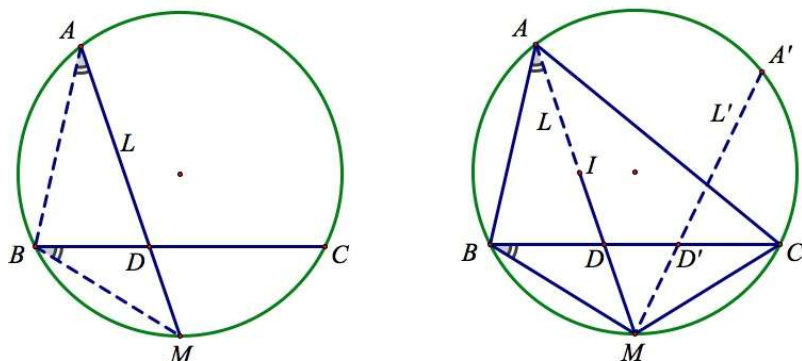
Let BC be a diameter of circle ω centred at O . Let A be a point of ω such that $\angle AOB < 120^\circ$. Let D be the midpoint of the arc AB which does not contain C . The line through O parallel to DA meets the line AC at I . The perpendicular bisector of OA meets ω at E and at F . Prove that I is the incentre of the triangle CEF .



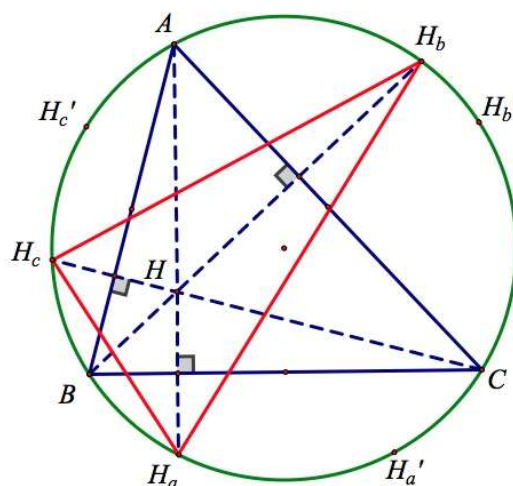
Shooting Lemma

Let M be the midpoint of arc BC of the circle ω . Let ray L emanating from M intersect segment BC at D and ω for the second time at A . Then:

- (a) $MD \cdot MA = MB^2$
- (b) If I is the incentre of triangle ABC , then $MD \cdot MA = MI^2$
- (c) If another ray L' from M intersects BC at D' and ω at A' , then $DD'A'A$ is cyclic.

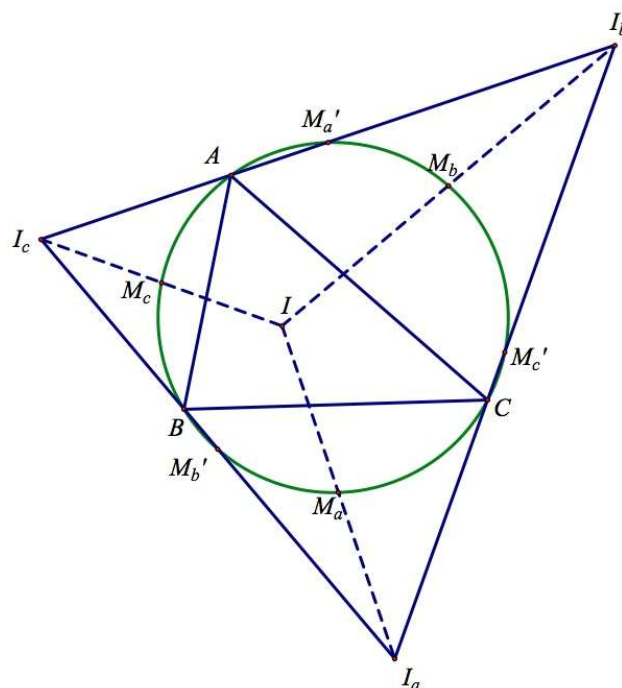
**Reflection of the orthocentre – revisited**

In the previous configuration, H is the incentre of triangle $H_aH_bH_c$.

**Excentres – the Big Picture**

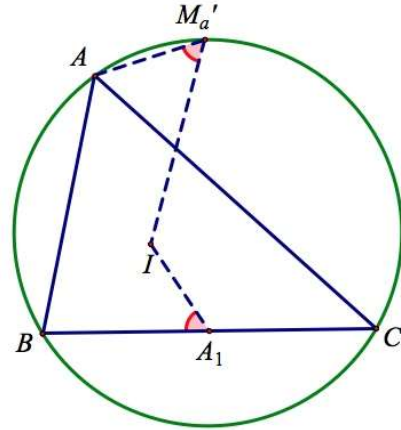
In triangle ABC with incentre I , let M_a, M_b, M_c the midpoints of the arcs BC, CA, AB that do not contain points A, B, C respectively. Further, denote by M_a', M_b', M_c' the antipodal points on the circumcircle of triangle ABC with respect to M_a, M_b, M_c respectively. Finally, let I_a, I_b, I_c be the excentres opposite to vertices A, B, C respectively. Then I is the orthocentre of triangle $I_aI_bI_c$ and the circumcircle of triangle ABC is the nine-point circle of triangle $I_aI_bI_c$. This has the following consequences:

- (a) Points M_a', M_b', M_c' are the midpoints of the respective sides in triangle $I_aI_bI_c$.
- (b) Quadrilaterals $BICI_a, CIAI_b, AIBI_c$ are cyclic with diameters II_a, II_b, II_c respectively. The centres of the respective circles are M_a, M_b, M_c .
- (c) Quadrilateral $I_bI_cBC, I_cI_aCA, I_aI_bAB$ are cyclic with diameters I_bI_c, I_cI_a, I_aI_b respectively. The centres of the respective circles are M_a', M_b', M_c' .

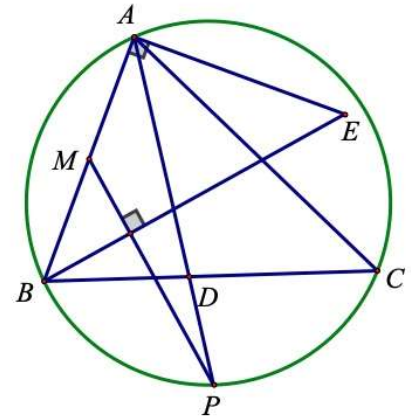


Example 3.4 (All-Russian Olympiad 2005)

Let ABC be a triangle and I its incentre. Denote by A_1 the midpoint of BC and by M_a' the midpoint of arc BC containing vertex A . Prove that $\angle IA_1B = \angle IM_a'A$.

**Exercise 1 (2020-21 October Test Problem 3)**

Let $\triangle ABC$ be an acute triangle with circumcircle S , let P be the midpoint of the minor arc BC of the circumcircle of $\triangle ABC$. Let AP meet BC at D , and let M be the midpoint of AB . Also, let E be the point such that $AE \perp AB$ and $BE \perp MP$. Prove that $AE = DE$.



Exercise 2 (CGMO 2021 P2)

As shown in the figure, let ABC be an acute angle with $AB < AC$, point I be its incentre, point J be its ex-centre with respect to A . Points X, Y are on the minor arc AB and minor arc AC of the circumcircle of triangle ABC respectively, such that $\angle AXI = \angle AYJ = 90^\circ$. Point K lies on the extension of BC beyond C such that $KI = KJ$. Show that the line AK bisects the segment XY .

